

NAINA OS -- VOLUME 8

MULTI-AGENT CIVILIZATION FRAMEWORK (MACF v1.0)

"One AI is useful. Hundreds of coordinated AI minds become an ecosystem."

DOCUMENT METADATA

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This document contains formal engineering specifications for the NAINA OS Agent Civilization Architecture, Executive Council, Agent Economy, Consensus Engine, The NAINA OS Constitution, and Final Platform Synthesis.

NAINA OS -- Multi-Agent Civilization Framework (MACF v1.0)

Volume 8: Executive Council, Agent Economy, Consensus Engine & The NAINA OS Constitution

One AI is useful. Hundreds of coordinated AI minds become an ecosystem.

Document Revision History

Date	Revision	Author	Description of Changes
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2026-07-31	v1.0.0	Chief Systems Architect	Release of Volume 8 -- Multi-Agent
2026-07-30	v0.9.0	Agent Ecosystem Team	Specifications for Executive Council

THE NAINA OS CONSTITUTION (Immutable Governing Document)

```
+-----+
|               THE NAINA OS CONSTITUTION               |
+-----+
| ARTICLE I: USER SOVEREIGNTY                             |
| The Human User maintains absolute ownership of data, memory, and code. |
+-----+
| ARTICLE II: LOCAL-FIRST PRIVACY                         |
| Zero telemetry or remote data transmission shall occur without explicit |
| capability token signature by the user.                 |
+-----+
| ARTICLE III: COGNITIVE SEPARATION RULE                  |
| NAINA (Companion) shall never execute code directly.    |
| CENANI (Operator) shall never generate unverified emotional dialogue.  |
+-----+
| ARTICLE IV: ADAPTER ISOLATION                          |
| No external engine or model shall couple directly to the kernel space. |
+-----+
| ARTICLE V: SECURITY VETO POWER                          |
| Security and Permission Agents possess absolute veto authority over     |
| any high-risk execution contract, regardless of agent consensus vote.   |
+-----+
```

DOCUMENT 08.1 -- Agent Civilization Architecture

NAINA OS transitions from a single assistant into a specialized, cooperative Society of AI Agents:



DOCUMENT 08.2 -- Executive Council Specification

The Executive Council coordinates complex multi-domain decisions before execution:

```
sequenceDiagram
    autonumber
    actor User
    participant NAINA as [NAINA] NAINA
    participant Council as [COUNCIL] Executive Council
    participant Security as [SEC] Security Agent
    participant FileSys as [HOST] FileSystem Agent
    participant Risk as [WARNING] Risk Analyzer

    User->>NAINA: "NAINA, purge all temporary build logs older than 7 days."
    NAINA->>Council: Convene Decision Session (Intent: PURGE_LOGS)

    par Multi-Agent Assessment
        Council->>Security: Verify Capability Token & Policy
        Council->>FileSys: Calculate Scope (10,000 files identified)
        Council->>Risk: Evaluate Impact & Recovery Path
    end

    Security-->>Council: Clearance Granted (CAP_FILE_WRITE active)
    FileSys-->>Council: Scope Payload (3.4 GB log files)
    Risk-->>Council: Impact Rating: HIGH (Mass Deletion)

    Council->>NAINA: Decision Contract (Requires User Clearance)
    NAINA-->>User: "The Executive Council identified 10,000 log files (3.4 GB). Proceed with delet"
```

DOCUMENT 08.3 -- Agent Society & Manifest Schema

Every agent in NAINA OS is instantiated as a managed service:

```
// Agent Society Manifest Schema (TypeScript)
export interface AgentManifest {
  agentId: string;           // e.g., "agent.coding.python"
  roleName: string;          // e.g., "Python Backend Developer"
  version: string;           // SemVer e.g., "1.2.0"
  ownerPersona: 'NAINA' | 'CENANI' | 'SYSTEM';
  assignedCapabilities: string[]; // e.g., ["CAP_FILE_READ", "CAP_EXEC_SHELL"]
  mcpToolsBound: string[];    // MCP tool bindings
  reputationScore: number;    // Dynamic score (0.0 to 1.0)
  resourceTokenLimit: number; // Max CPU/VRAM token allocation
}
```

DOCUMENT 08.4 -- Agent Economy & Resource Tokenization

To prevent any single agent from monopolizing system hardware, NKRS implements an Internal Token Economy:

- * Compute Tokens (cTokens): Measure CPU cycles, GPU VRAM, and RAM memory usage.
- * Resource Allocation: High-priority tasks (Voice STT/TTS) receive guaranteed token pools; background agents (Obsidian Vault Indexer) operate on idle token budgets.

DOCUMENT 08.5 -- Agent Reputation System

Agents are continuously evaluated across six operational metrics:

$$\text{Reputation Score} = 0.30(\text{Success Rate}) + 0.20(\text{Accuracy}) + 0.20(\text{Reliability}) - 0.15(\text{Latency}) - 0.15(\text{Resource Cost})$$

Unreliable agents (Reputation < 0.50) are automatically demoted by the Router in favor of backup agents.

DOCUMENT 08.6 -- Collective Memory Architecture

All domain agents contribute to and query a unified Collective Memory Core:

```
[Specialized Agents] --> [Shared Event Bus] --> [Knowledge Graph & pgvector] --> [Obsidian Vault]
```

DOCUMENT 08.7 -- Agent Collaboration Protocol

Web Development Task Pipeline Example:

```
[User Request] --> [Planner Agent] --> [Research Agent] --> [UI Designer Agent]
                                     |
[Done] <-- [Review Agent] <-- [Deployment Agent] <-- [Frontend/Backend Agents]
```

DOCUMENT 08.8 -- Swarm Intelligence Engine

The Swarm Intelligence Engine orchestrates 100+ micro-agents executing in parallel (e.g., parallel code auditing, multi-site research, multi-file image metadata extraction). Results are merged by the Master Orchestrator.

DOCUMENT 08.9 -- Conflict Resolution & Consensus Engine

When domain agents disagree (e.g., Research = YES, Security = NO, Planner = MAYBE), the Consensus Engine evaluates votes:

- * Veto Authority: Security and Permission agents hold absolute veto authority over destructive actions.
- * Weighted Consensus: Votes are weighted by agent reputation scores.

DOCUMENT 08.10 -- Agent Evolution Lifecycle

```
[New Agent Created] --> [Register Manifest] --> [Pass Automated Test Suite]
                                     |
[Ready for Production] <-- [Executive Approval] <-- [Capability Token Grant]
```

THE FINAL PLATFORM SYNTHESIS

NAINA OS is the realization of an open-source, local-first, memory-centric AI Operating Platform:

```

                                THE NAINA OS PLATFORM
+-----+
| [NAINA] NAINA: Digital Human Companion, High-EQ Interface & Goal Planner |
| [CENANI] CENANI: Systems Operator, Infrastructure Automation & Engineering |
| [COUNCIL] EXECUTIVE COUNCIL: Multi-Agent Governance & Risk Consensus      |
| [AI] MKIE MEMORY: pgvector, 16-Entity Knowledge Graph & Obsidian Vault    |
| [ENGINE] NKRS KERNEL: Microkernel Event Bus, CBAC Tokens & Docker Sandbox |
| ? ADAPTER MESH: ARAL Interface, OpenClaw Engine & Hermes Model Router     |
| [CONSTITUTION] CONSTITUTION: Sovereign User Ownership, Privacy & Security Veto |
+-----+
```

End of NAINA OS Multi-Agent Civilization Framework -- Volume 8 (Documents 08.1 - 08.10 & The Constitution)